



Fairness and ethics (and debates!)

Dr. Dina Strikovic | Meeting 4





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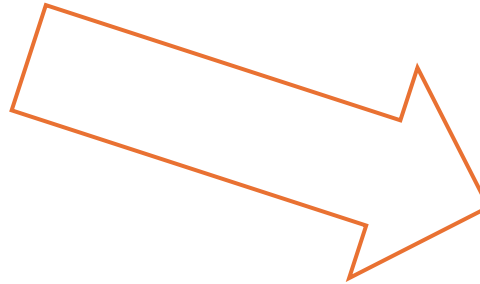




Recap: Datafication, Bias, NRS

Three topics that we focused on

- Datafication
- Bias
- News Recommender Systems



- Who makes the decisions?
- And what do they base those decisions on?



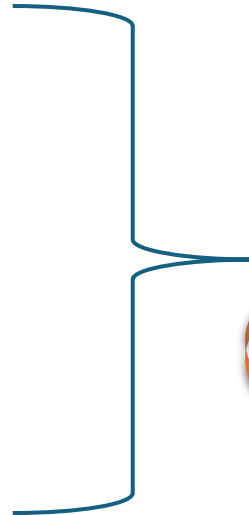
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Recommender Systems

Ethics and Fairness

What is "ethics"?

- Recommenders can be deployed in contexts that are morally loaded

- Health care
- Information distribution
- Lifestyle
- Insurance
- Labour market
- Social connections

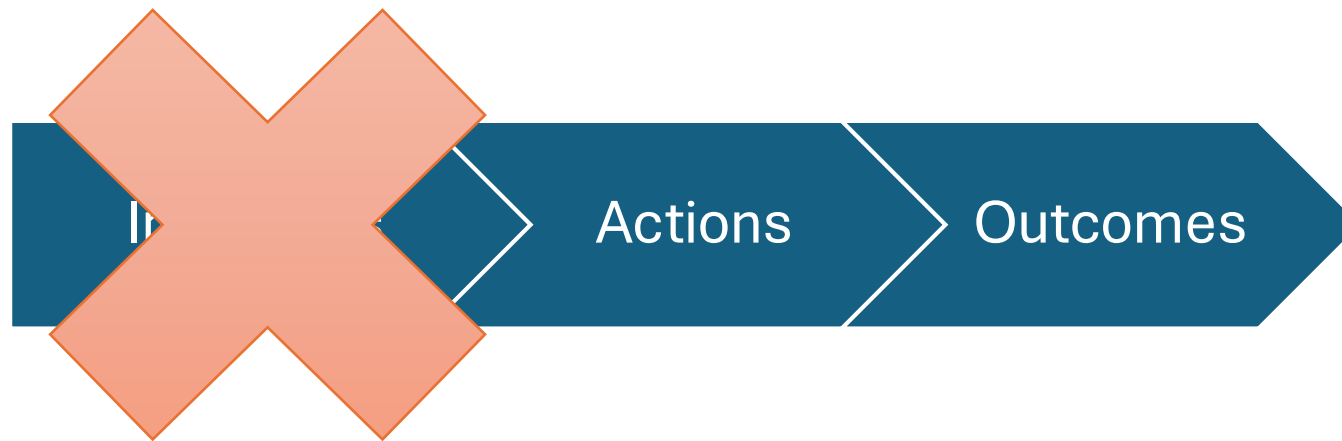


Different contexts and fields – how can we agree on what the **right thing to do** is?

Which moral principles are correct?

What is "ethics"?

- Moral principles and what is right = deeply contentious and debated in philosophy
- BUT we don't have to take a side (and certainly not on all questions)
- There are some questions that are more relevant to recommenders than others





What is "ethics"?

In terms of Action and Consequence, then, a recommender can:

- Impact (negatively) the utility of any of its stakeholders
- Violate their rights

These can be...

- ...immediate (recommendation is incorrect and not useful)
- ..., or pose future risks.

Inappropriate Content

- Recommended and collected

Privacy

- RS depends on user information: what is the trade-off?

Autonomy and personal identity

- Addiction economy: manipulative interventions; RS as sticky traps

Opacity

- Transparency, but make it make sense

Social effects

- Filter bubbles, diversity, etc.

Fairness

- Complicated, multi-sided concept



Ethical impact assessment framework

“An ethical impact assessment must not only focus on the ethics of a technology, but on the technology itself, its *values*, how it is *perceived* and how it is *used* or *might be used* in the future, not only by itself but as *a component in a larger technological framework*.” (Wright, 2011, p. 204)



Some ethical principles

- Privacy and data protection
- Respect for autonomy (right to liberty)
- Nonmaleficence (avoiding harm)
- Beneficence (doing good!)
- Justice



Some ethical principles

- Privacy and data protection
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- Collection limitation (data minimization) and retention
- Data quality
- Purpose specification
- Use limitation
- Confidentiality, security, and protection of data
- Transparency (openness)
- Individual participation and access to data
- Anonymity
- Privacy of personal communications: monitoring and location tracking
- Privacy of the person
- Privacy of personal behaviour



Some ethical principles

- Privacy and data protection
- Respect for autonomy (right to liberty)
- Nonmaleficence (avoiding harm)
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- Free from control
- Free to accomplish the things we want ("Stolen Focus")
- Informed consent
- RS can be trained to identify psychological vulnerabilities
- User responsibility: regulation



Some ethical principles

- Privacy and data protection
- Respect for autonomy (right to liberty)
- **Nonmaleficence (avoiding harm)**
- Beneficence (doing good!)
- Justice

- Safety and harassment
- Social solidarity, inclusion and exclusion
- Isolation and substitution of human contact
- Discrimination and social sorting



Some ethical principles

- Privacy and data protection
- Respect for autonomy (right to liberty)
- Nonmaleficence (avoiding harm)
- **Beneficence (doing good!)**
- Justice

- Universal service: inclusivity, etc.
- Accessibility
- Value sensitive design: ethics check at all points
- Sustainability: are people going to be OK without the service?



Some ethical principles

- Privacy and data protection
- Respect for autonomy (right to liberty)
- Nonmaleficence (avoiding harm)
- Beneficence (doing good!)
- **Justice**

- Equality
- Fairness: but, complicated discussion – what does this mean? How do we ensure it? Who measures it? How?
- No clear parameters for "fairness"



What is "fairness"?

Human value as algorithmic goal

- *Group vs Individual*: equality for individuals/groups
 - Candidates with similar qualifications should be ranked equally for jobs
 - Separate groups have similar experience/ benefits in decision making, for example
- *Direct vs. Indirect*: preferential treatment based on direct or indirect (proxy) characteristics
- *Static vs. Dynamic*: is the recommender fixed or does it take user interactions into account?
- *Short-term vs. Long term*: what is the end goal?



Fairness for whom?

Recommenders are multi-sided projects

- Different stakeholders + platform itself = affected by (un)fairness
- Consumer vs. Provider viewpoints
 - Differs by platform: what responsibilities do we have towards our users
 - News media vs. Social media --> different responsibilities for the balance of user *wants* and *needs*
 - Still: with power comes responsibility!
 - Intersectionality: Individual/group fairness cannot be measured on one demographic...

Measuring fairness

- Output: distributive fairness – fairness of decision outcomes

But... different formal definitions

- **Equality:** equal distribution of outcomes regardless of individual contributions and preconditions
--> every person has the same chance to receive their preferences
 - **Equity:** outcomes that are perceived as just when they are allocated corresponding to individual contributions *--> Those who contribute more, deserve more.*
 - **Need:** outcomes are distributed according to individual situations and preconditions *--> those who need the resources most get preferential treatment.*
- Fairness of algorithmic processes/procedural fairness



Fairness and ethics: any guidelines?

DEDA: Data Ethics Decision Aid

- Specifically, when thinking about value-sensitive design: concludes that ethics is **not a binary decision** but in every step of the design.
- Sample questions →

The image shows a vertical form titled 'COLLECTION' on the left side. It is divided into two main sections: 'ALGORITHMS' and 'SOURCE'. Each section contains a list of questions, each with a red circular number next to it. The 'ALGORITHMS' section contains questions 8, 9, and 10. The 'SOURCE' section contains questions 11, 12, and 13. The form is light blue with a white background for the text.

Section	Question	Number
ALGORITHMS	Does this project use an algorithm, or some form of machine learning or neural networks (AI)? If not, go to 'Source'.	8
	Can someone on the team explain how the algorithm in question works?	9
	Can someone on the team explain the algorithm to the public in a way that is understandable?	10
SOURCE	Where does the data come from?	11
	In what ways have you checked the quality of the data?	12
	Is the data only relevant up to a certain point in time? Is there an "expiration date"?	13

Fairness and ethics: any guidelines?

DEDA: Data Ethics Decision Aid

- Specifically, when thinking about value-sensitive design: concludes that ethics is **not a binary decision** but in every step of the design.
- Sample questions 

TRANSPARENCY

29

Does the project risk generating public concern or outrage?

30

How transparent are you about this project towards citizens?

31

Do citizens have the opportunity to raise objections to the results of the project?

32

Can citizens opt out of their involvement in the project?
If so, when and how?

Fairness and ethics: any guidelines?

DEDA: Data Ethics Decision Aid

- Specifically, when thinking about value-sensitive design: concludes that ethics is **not a binary decision** but in every step of the design.
- Sample questions 

BIAS 

37

As a member of the project, what outcomes do you expect?

38

Is there any part of the project that you consider problematic? If so, discuss your concern.

39

Will the results of the analysis be evaluated by a human before being implemented?

40

Is there a risk that the project could contribute to discrimination against certain people or groups?

41

Are all relevant citizens adequately represented within your data? Which citizens are missing, over-represented or under-represented?

42

Does your model have a feedback loop that could create negative consequences?

43

Are you gathering the information that is appropriate for the purposes of your project?

44

Function creep: can you imagine a future scenario in which the results of your project are (mis)used for alternative purposes?

45

Do your answers to these questions change when you consider possible long-term effects? Why?



Fairness and ethics: any guidelines?



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Part of [Chapter IV: Transparency Obligations for Providers and Deployers of Certain AI Systems](#)

Article 50: Transparency Obligations for Providers and Deployers of Certain AI Systems

Date of entry into force: **2 August 2026**
According to: **Article 113**

See here for a [full implementation timeline](#).

SUMMARY +

1. Providers shall ensure that AI systems intended to interact directly with natural persons are designed and developed in such a way that the natural persons concerned are informed that they are interacting with an AI system, unless this is obvious from the point of view of a natural person who is reasonably well-informed, observant and circumspect, taking into account the circumstances and the context of use. This obligation shall not apply to AI systems authorised by law to detect, prevent, investigate or prosecute criminal offences, subject to appropriate safeguards for the rights and freedoms of third parties, unless those systems are available for the public to report a criminal offence. [Related: Recital 132](#)
2. Providers of AI systems, including general-purpose AI systems, generating synthetic audio, image, video or text content, shall ensure that the outputs of the AI system are marked in a machine-readable format and



EU AI Act

Proposal for a
Regulation of the European Parliament and of
the Council Laying Down Harmonised Rules on
Artificial Intelligence (Artificial Intelligence Act)
and Amending Certain Union Legislative Acts

2021/0106 (COD)

European
Commission



Fairness and ethics: any guidelines?

- Ethical principles must guide the assessment of large and small decisions *before, during* and *after* their implementation
- Privacy is important, but not the *only* ethical principle
- The rest is (largely) up to you!



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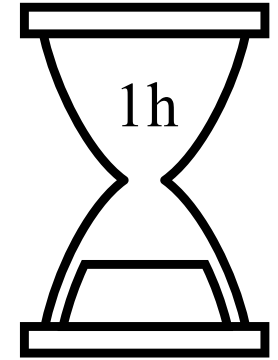
Debates!



Assignment

6 Groups/3 Debates

- Each group gets a scenario: NRS, Social Media or Hiring Algorithm
 - 2 Groups per scenario
- You get a hypothetical proposal for an implementation of a recommender system
- Each group is assigned a position to argue for: either SUPPORT or OPPOSE
- DO NOT argue your personal opinion!
- DO: justify your assigned position using course concepts and literature, articulate the trade-offs of your approach, and critically evaluate the risks and limitations of the alternative design



Procedure

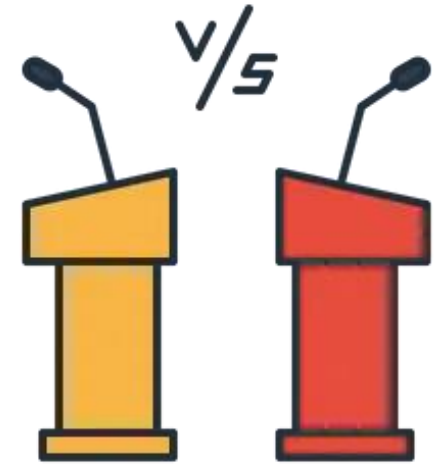
- Opening statements (2 minutes each)
 - Each team presents its argument, providing justifications rooted in course materials
 - Do NOT respond to the opposing team, yet!

[Re-group: 3 mins]

- Open debate (5-10 minutes)
 - Incl. questions from opposing team and jury focusing on technical, ethical, and societal implications

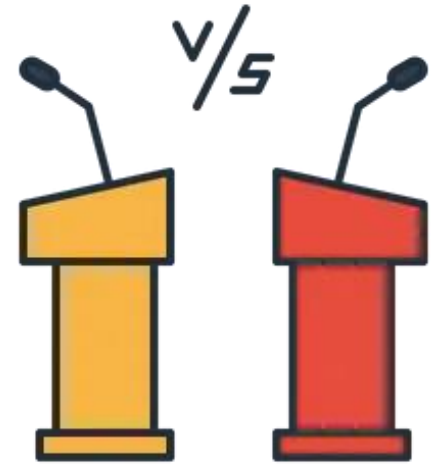
[Regroup: 3 mins]

- Closing statements (2 minutes)
 - Break: Debating teams leave the room and “Jury” discusses (~5 minutes)



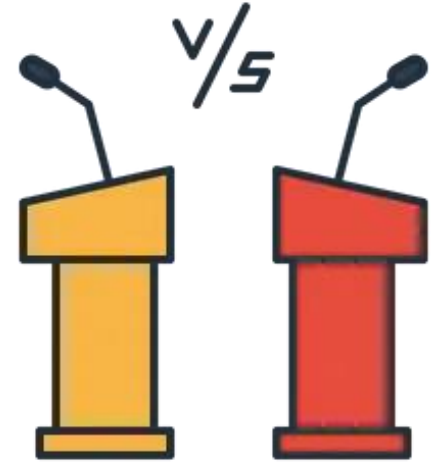
Ground rules

- No interruptions during opening and closing statements
- Be respectful! The goal is an informed debate, not to clip “got-cha” moments
- Try not to get bogged down in the legal or technical details, but focus on the ethics elements
- Criticize ideas, not people...
- **No recording** of any kind



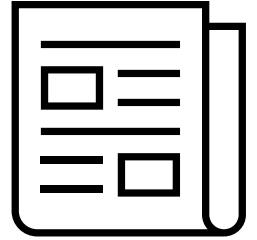
Task divisions

- Moderator?
- Time keeper?
- Jury:
 - Evaluate the arguments *as presented*
 - Avoid your personal opinions/perspectives, or other arguments that are not presented
 - Evaluate tone, professionalism as well
 - Who made the strongest case for their position? How so?





Groups 1&2: News recommender system



NRS

Many major news outlets have recommender systems that look at readers' past behaviour to create a personalized recommendation. This has turned out to work really well to engage users, but only in terms of the suggested topics.

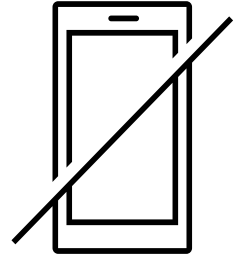
HYPOTHETICAL:

The marketing department found that users leave to different platforms to get more diverse news. Other news outlets found the same thing. There is now a proposition:

A collaboration with all major Dutch news outlets to offer all news through one single recommender system. Users would subscribe for a small fee, and in return, have access to all the major news outlets on the same news app. The outlets are now paid a small, equal share of the subscription fees, plus bonuses for clicks on their own outlets' articles.”



Groups 3&4: TikTok Proxies



TikTok Proxies

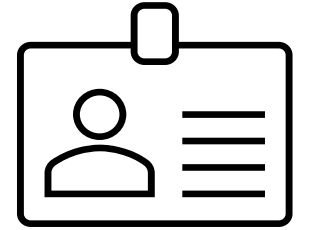
TikTok has made special efforts to remove accounts belonging to users under the age of 13 in Europe. Other platforms are facing similar challenges and discussions. One major downside of age restrictions is that users attempt to circumvent them by posing as older than 13 (for instance, by falsifying ID verification).

HYPOTHETICAL: EU regulators suggest that TikTok and other platforms that rely on age verification should use proxies to detect whether a user is under 13, in order to ensure that age regulations are accurately implemented.

This would mean that these platforms will get access to data that is not collected on the platform itself but is legally shared by other platforms and data collectors. In this way, they could more accurately predict or verify users' ages and better protect children under 13 from accessing the platform.



Groups 5&6: Hiring algorithms & risk assessment



Hiring Algorithms

Many jobs in the security sector require a thorough background check - including information about family background and ideology, the people in candidates' social circles, and the organizations or clubs candidates are or were part of, among other factors. Currently, this process is conducted by human investigators. It can take several months. This slows down hiring processes and exposes judgments about political ideology, background, and social standing to potential human bias.

HYPOTHETICAL: Hiring managers propose implementing algorithmic screening systems to support this process. The system would have access to these candidate data and conduct an initial ranking of suitable applicants, predicting potential security risks associated with each candidate.

To prevent biases from entering the algorithm, the proposal includes applying an automatic “context correction factor”: applicants from underprivileged demographic strata would receive a systematic score adjustment to compensate for structural disadvantages, even if this reduces the system's predictive accuracy. The aim is to improve efficiency while also proactively addressing bias and promoting social mobility and fairness in the hiring process.